Welcome to COS30049: Computing Technology Innovation Project

Duration:	One teaching period
Contact hours:	Recommended 12 hours of study per week
Pre-requisites	Nil
Credit points	12.5

Aim

This is a project-based unit where students collaborate in teams to develop innovative solutions for industry challenges, with a particular emphasis on solving real-world cybersecurity problems. The primary focus of this project is to apply a range of IT techniques and analysis methods to effectively address these challenges. Throughout the projects, student teams will be supported by a designated staff member acting as a facilitator. The aim is to guide the teams in creating automated tools or platforms, providing instructions that cater to diverse ICT backgrounds. Students will learn how to complete a real-world project from the ground up, just like what industry professionals do every day in their companies.

Unit learning outcomes (ULOs)

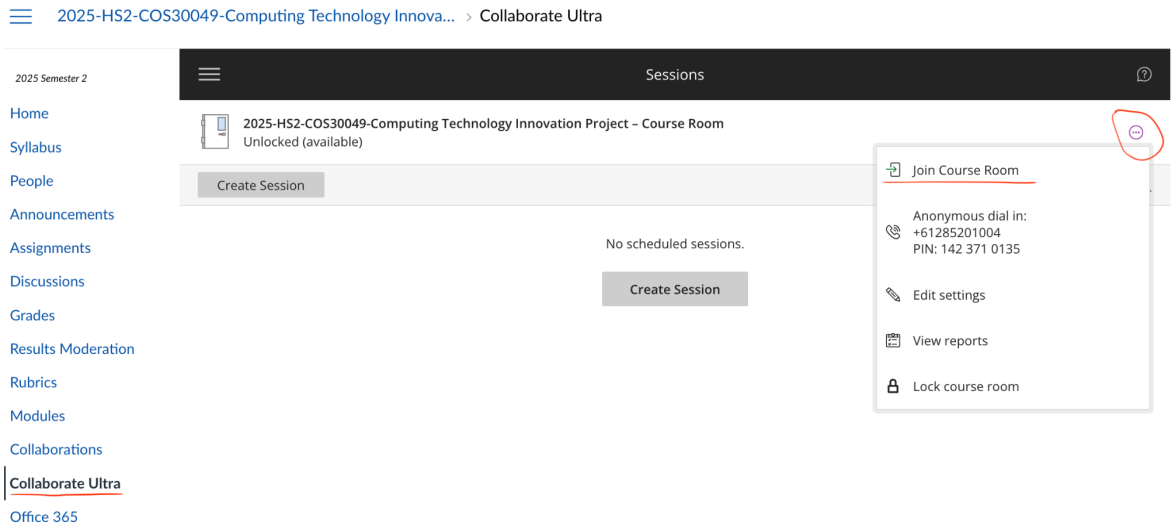
- ULO1: Apply a systematic approach to computing technology innovation.
- ULO2: Apply knowledge of innovation fundamentals to computing technology challenges.
- ULO3: Find, organise, and make decisions on a range of topics related to computing technology innovation.
- ULO4: Use technology to develop and present innovation solutions.
- ULO5: Demonstrate reflective practice and use self and peer evaluation.
- ULO6: Communicate within teams and stakeholders using appropriate verbal, written and technological approaches.

This unit may contribute to the development of the following Swinburne Graduate Attributes:

- GA1: Communication 1 - Verbal Communication.
- GA2: Communication 2 - Communicating using different media.
- GA3: Teamwork 1 - Collaboration and negotiation.
- GA4: Teamwork 2 - Teamwork roles and processes.
- GA5: Digital Literacies 1 - Information literacy.
- GA6: Digital Literacies 2 - Technical literacy.

Live Lecture

Here is the guideline to get access to the live lecture. We will use Collaborative Ultra.



Team Setup

This unit is a large team-based course that brings together students with diverse backgrounds.

Every student must register for one group from a predefined set, with each group comprising three students from the same workshop session. During the first week of the semester, students can autonomously select the group they wish to join. By no later than Friday of Week 2, all students must be part of a group. If any student fails to join a group within this timeframe, the teaching team will step in and randomly assign them to a group over the weekend following Week 2 (Saturday and Sunday).

To foster a cohesive team environment where each member is familiar with one another, we strongly encourage students to take the initiative to self-organize into teams of three within their own session and then join a group together. To facilitate this process, we have outlined the following steps:

- Go to the People section on Canvas.
- Select a group from the list that best aligns with your preferences.
- Click to confirm your choice.
- Note: After submitting Assignment 1, group changes will no longer be allowed.

Content & Assignment

Throughout this unit, we will focus on applying AI technology as the central theme for our cybersecurity-related projects. Students will choose their topic (must be cybersecurity-related) from a selection of provided options and use the AI technology they learn during the semester to develop a website that showcases their chosen topic. Additionally, students will apply project management techniques to manage their work effectively, mirroring the practices of industry professionals.

The unit will guide students in completing a real-world cybersecurity project from the ground up, even with no prior background, emulating the experiences of industrial professionals. The coursework is structured around three distinct projects.

Project Assignments

Assignment 1: IT Project Management Plan Report

Weight: 15%
Description: Students will create a comprehensive project management plan, detailing the scope, timeline, resources, and risk management strategies for their AI-based project.

Assignment 2: Trained AI Model Tailored for a Specific Cybersecurity-related Topic

Weight: 40%
Description: Students will develop and train an AI model tailored to their chosen cybersecurity-related topic. This assignment will assess their understanding and application of AI techniques and tools covered in the course.

Assignment 3: Dynamic Website to Present the AI Model that Addresses Real-world Cybersecurity Challenges

Weight: 45%
Description: Students will design and develop a dynamic website to present their AI model and its outcomes on the cybersecurity-related topic.

By completing these cybersecurity-related projects, students will gain hands-on experience in AI development and project management, preparing them for real-world cybersecurity-related challenges in the tech industry. The due dates for each group set are presented in the table below:

Assignment task	Time limit (+/- 10%)	Individual/team task	Weighting	Due date
Assignment 1	week2~4	Teamwork	15%	Sunday AEST 11:59 pm Week 4
Assignment 2	week5~8	Teamwork	40%	Sunday AEST 11:59 pm Week 8
Assignment 3	week9~13	Teamwork	45%	Sunday AEST 11:59 pm Week 13

Minimum requirements to pass this unit of study

To pass this unit, you must:

(i) achieve an overall mark for the unit of 50% or more, and
(ii) complete each of the three projects to an acceptable standard (at least 40% of each assignment).

A rubric will be used to determine if students have met the acceptable standard. Students will not meet the hurdle requirements if any one of the three projects is not completed.

Referencing

To avoid plagiarism, you are required to provide a reference whenever you include information from other sources in your work. You can check your assignments ahead of submission by using a program called Turnitin. The Academic practice page in the Student Hub has information on plagiarism, referencing and Turnitin.